Lab #3: Thermal Radiation & the Statistics of Noise

Setting up your laptop for the AirSpy

For lab #3, you'll be required to record data using an AirSpy R2 software-defined radio (SDR) USB device. We have a collection of old / disappointing departmental laptops which can be used for this purpose, but you're welcome to use your own computer by installing the control software. This handout gives some tips on how to get it running.

Windows

If you're running a windows laptop, you can run the airspy through the Windows Linux Subsystem, it's mostly straightforward though takes a bit of effort to give USB access.

1. Install the WSL, as per Microsoft instructions:
    a. Open PowerShell in Administrator mode and run **`wsl --install`**
2. Open a WSL terminal, install airspy:
    a. **`sudo apt-get update`**
    b. **`sudo apt-get install airspy`**
3. Give WSL access to the USB device
    a. Install USBipd
    b. Go back to your administrator PowerShell
    c. Find the right USB by running **`usbipd list`**
    d. Connect the airspy to WSL with **`usbipd bind --busid X-Y`** (with appropriate values for X and Y taken from the listing above.)
    e. Some people have had to use **`usbipd attach -w --busid X-Y`**
4. Try it out, in the same WSL terminal, run **`airspy_info`**
5. Depending on your WSL install, the default user may not have permission! Run with full privileges but adding **`sudo`** before each command, e.g. **`sudo airspy_info`**

MacOS

The airspy software is straightforward to install under MacOS via the Homebrew package manager.

1. Install Homebrew following the instructions on their webpage.
2. Open a terminal, and run **`brew install airspy`**
3. Try it out, in the same terminal, run **`airspy_info`**

Ubuntu / Debian Linux variants

This one's typically easiest.

1. Open a terminal, run **`sudo apt-get install airspy`**
2. Try it out, in the same terminal, run **`airspy_info`**
3. On the off chance your user doesn't have access to the USB, try **`sudo`**'ing it.